

Hoang Minh Thanh

 [hmthanh.github.io](https://github.com/hmthanh)  [hmthanh](#)  [hmthanh](#)  Ho Chi Minh City, Vietnam
 hmthanhgm@gmail.com  OCID  Google scholar

Passionate about Digital Humans, WebGL, and real-time rendering. I combine a strong foundation in machine learning with hands-on expertise in 3D human modeling. My contributions to DeepGesture and the GENE Challenge demonstrate capabilities, from training transformer and diffusion models to deploying evaluation pipelines and interactive 3D systems using Python, C++, and Web technologies. I aim to bring both research rigor and engineering execution to advance research development.

Keypoint

- **Comprehensive knowledge about DeepLearning:** [DeepGesture](#), [GraphAttention](#), [SurveyWithCode](#)
- **Solid computer graphics and Practical ML systems:** [Fillament remake](#), [My custom facial render engine](#), experiment built end-to-end ML systems from data preprocessing and model training.
- **Passion about Digital Human and Human Computer Interaction:** Creator of [OpenHuman](#), [Cortexpod](#)
- **Collaborative Research:** Co-organizer of gesture generation [GENEA Leaderboard](#). Worked with international research teams and contributed to reproducible research artifacts supporting academic advancement.

Education

AS	University of Science - VNUHCM, Information Technology	2012 – 2015
BS	University of Science - VNUHCM, Computer Science	Sept. 2018 – Sept. 2020
	• GPA: 3.3/4.0 • Thesis: GCAT - Link Prediction in Knowledge Graph [Report] [Code] • Advisor: PhD. Le Ngoc Thanh	
MSc	University of Science - VNUHCM, Computer Science	Oct. 2021 – Dec. 2024
	• GPA: 3.44/4.0 • Thesis: OpenHuman: A conversational gesture synthesis system based on emotions and semantics [Report] [Code] • Advisor: Assoc. Prof. Dr. Ly Quoc Ngoc	

Publications

DeepGesture: A conversational gesture synthesis system based on emotions and semantics [arXiv] [homepage]	Jul. 2025
Towards a GENE Leaderboard [arXiv] [homepage]	Oct. 2024
Graph Collaborative Attention Network for Link Prediction in Knowledge Graphs [arXiv] [homepage]	Oct. 2020

Experience

Rakumo Inc, Software Engineer	Jul. 2019 – Apr. 2020
• Developed a Django-based web application with integrated DocuSign.	
FPT Software, AWS Data Engineer	Oct. 2020 – Nov. 2021
• CI/CD data pipeline terraform in Jenkins, implemented and modified real-time and batch data pipelines on AWS using Kinesis, Lambda, S3, Glue, StepFunction, SNS, ..etc..	

- Cox Automotive (onsite-offshore): Collaborated with international teams in Agile

VNG Corp, Software Engineer

Oct. 2021 – May 2024

ZDN Team (Zalo Content Delivery Network): caching service handle millions req

- Implement scheduled threadpool for monitoring stats execution.
- Integrate outlier detection for migrating DDoS within the caching server.
- Develop dashboard to real-time update server status.

OpenHuman (openhuman.ai), Creator

May 2024 – Present

- Join GENE Research Team ([GENEA Leaderboard](#), [HEMVIP](#)): Designed and built a full-stack, production-ready leaderboard system to evaluate multiple DeepLearning gesture generation models, integrating user study workflows via Prolific.

Projects

DeepGesture ([Code](#)): Diffusion model for generative human motion with Transformer-based. Integrate with DeepPhase-Unity. deepgesture.pages.dev

Knowledge Graph Attention Network ([Code](#)): Performs link prediction on knowledge graphs by training TransE models with an attention mechanism applied to n-hop neighborhood graphs. graphattention.github.io

MillionScope ([Code](#)): ChatGPT-like with real-time streaming responses millionscope.com

SurveyWithCode: PapersWithCode alternative surveywithcode.dev

Web-base realisitic human render ([Code](#)): load mesh, apply glsl material: shine, coat, roughness, reflection, diffuse map. renderengine.pages.dev

Pet projects:

- [Pucca Runner](#): pathfinding game Dijkstra's algorithm
- [Blendshape animation](#), [MetaPet](#), [cortexpod.com](#); [pithagon.com](#)

Certifications

- [Critical Thinking Certification 2014](#), [Emotion Recognition Certification 2019](#), [Summer ML](#)
- Coursera: [Design Pattern](#), [TensorFlow Developer Professional](#) 2021
- **AWS**: [Solutions Architect - Associate](#), [Machine Learning - Specialty](#) 2021
- [Solution Architect Certifications](#) 2021

Technologies

Expertise: transformation matrix, digital human, quaternion, representation learning, animation synthesis, real-time scalable system

Proficient: Python, Jupyter notebook, Web (React, Next.js, Vite, mdx.js, tailwindcss), WebGL, GLSL Shader.

Tools: MotionBuilder, Maya (blendshape, rigging), Blender (bones, shapekey, hair particle), Zbrush (Projection, Layers), RDS ZWrap (Transfer topology), Marmoset Toolbag, Unity, Photoshop

Experience: BRDF, Retarget animation, transfer facial expression, skeleton builder, photogrammetry